Interior Column Removal for Hall A and Hall B: Research and Sanity Check

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Property: 1699 Sycamore View Rd, Memphis, TN (Shelby County), also listed as 1735 Sycamore View Road: a single-story building of 127,634 SF, built in 1984 and renovated in 1996.
Prepared: October 18, 2026
Question: Can the 4 interior roof columns in the two banquet rooms (Hall A and Hall B, 131.8 ft x 83.5 ft each) be removed? What would it cost, how long would it take, and what are the risks?

How to read this: Every researched figure has a link. My own calculations and assumptions are labeled **[calc]** or **[assumption]**. Where I couldn't find something, I say so. This is a budgeting and feasibility document. It does not replace design by a Tennessee-licensed structural engineer.

1. Bottom line

Question	Answer
Is this routine?	**Yes. It is an engineered retrofit, not a minor one.** Big-box, mall-anchor and factory conversions regularly remove interior columns to create worship, rink and recreation spaces, using new trusses or transfer girders. In every precedent below, a structural engineer designed the work and the job included temporary shoring or jacking. Most also included column or footing work.
Cost per removed column (all-in)	**Low ≈ $105k · Mid ≈ $275k · High ≈ $495k** (build-up in §3.8)
All 4 columns	**Low ≈ $0.42M · Mid ≈ $1.10M · High ≈ $1.98M**
Feasibility budget (ownership group hires the steel fabricator, erector and structural engineer directly; Memphis prices)	**Low ≈ $85k · Mid ≈ $175k · High ≈ $300k per column; ≈ $0.34M / $0.70M / $1.20M for four.** This is the build-up below without the general contractor's 12–18% for general conditions and overhead and profit, with direct costs at Memphis prices (RSMeans 2025 city cost index 89.4), and with engineering, inspection and the contingency kept.
Phase 1: Hall A only (2 columns)	**Low ≈ $0.21M · Mid ≈ $0.55M · High ≈ $0.99M.** Add about 5–10% when Hall B follows later, for a second mobilization **[assumption]**.
Schedule	About **6–9 months** from hiring the engineer to columns out, at typical 2026 steel lead times. **12+ months** if steel runs at the 40–50 weeks Skanska reports in some markets. Field work is about 2–3 weeks per column, or 6–10 weeks for all four.
Top 3 design risks	(1) The end columns and footings pick up about **60–70% more load**. (2) An 89-ft roof girder is governed by **stiffness and ponding**, so it will be deep and will cost headroom. (3) The change to assembly (A-2) occupancy can trigger **code upgrades**: Risk Category III, which leads to the IEBC seismic requirements in Memphis.
Recommendation	**Budget about $175k per column, about $0.7M for four, with the work bid trade by trade (contractor-delivered, the Mid case is about $275k per column and $1.1M for four), with Hall A first.** Before any binding commitment, get a one-day structural walkthrough (§8) to confirm the framing type and direction, clear height, roof slope and drainage, and column and footing sizes. Raise the Risk Category / IEBC question at the City pre-application meeting.

2. Is this routine? Precedents

Project	Former use	What was done	Structural method (as the source states it)
Tulsa WeStreet Ice Center, Tulsa, OK	Empty mall-anchor department store	**36 columns removed**; 18 of them supported the joist and joist-girder roof. Bays went from 28×56 ft to 28×112 ft.	**Six new steel trusses span 112 ft.** Each is a pair of inverted bowstring trusses, one on each side of the existing columns and joist girders. The existing columns serve as vertical web members, the top chords are grout-filled HSS, and the bottom chords are 2–3 in. high-strength rods sized by deflection. **18 footings replaced**, 2 columns reinforced, and trusses **jacked 2 to 3½ in.** to set camber. **510 tons** of renovation steel in total. Opened about 18 months after kickoff. (Modern Steel Construction, "Retail to Rinks," June 2024)
Christ Church of Northwest Arkansas	Former Sears anchor at the NW Arkansas Mall, later retrofitted as a call center	"Ten columns were removed and replaced with new structural members" to open an 850-seat worship center on a 24-ft grid. Opened May 2024 after an accelerated design phase.	System not named; the source mentions concealed beams. (MBL Architects)
Crossroads Church, Florence, KY	Former big-box retail store (100,000 SF)	**3 columns removed** for sightlines in a 1,600-seat sanctuary; roof raised within four bays.	Not named. (Champlin Architecture)
Eagle Brook Church, Anoka, MN	Vacant Kmart (~60,000 SF)	1,000-seat auditorium: "To minimize columns within the worship space, the existing structure was converted to long-span." Completed 2017.	Long-span conversion; materials not named. (BWBR)
Chelsea Piers Connecticut, Stamford	Former factory (400,000 SF)	**23 columns removed** for 100-ft column-free zones.	**King-post trusses built from the in-place roof structure.** The existing beams became the compression chords and some columns were kept as king posts. "Only a relatively small amount of steel had to be added," as steel-angle tension chords. (WSP; see also Club Solutions)
The Heights Church, Henrico, VA	Old shopping center (40,000 SF project)	Existing columns removed for a 1,500-seat sanctuary; roof raised 10 ft.	"Suspending the structure from new steel trusses." (L.F. Jennings)
Iglesia Cristiana Segadores de Vida, Miami	Existing 200,000 SF building	"Almost all of the existing columns are removed" for a 10,000-seat sanctuary; roof height nearly doubled.	"Deep, long-span custom steel trusses spanning 150 feet." (BNI Engineers)

Project	Former use	What was done	Structural method (as the source states it)
Legacy Church, La Mesa, CA	14,500 SF strip-mall bakery and bar	Sanctuary cleared of columns, about 425 seats.	"New steel jack beam assemblies." (BGW Services)
Waters Church, N. Attleboro, MA	Former plant	Second floor removed for the sanctuary; roof re-supported.	A steel truss "five-and-a-half feet-tall, by eighty-feet long", plus new footings and perimeter columns. (Worship Facilities, May/June 2017)
Crossgates Mall, Albany, NY (2014)	Base of a 4-story building	**2 columns removed** for a future auditorium; new span about 75 ft carrying 1,000 kips.	The 5-ft-deep steel beam option was priced at $957k–$963k and rejected. A post-tensioned concrete beam was built for $413k. "3 weeks to design, 2 months to implement in the field." (ProCM case study)

Practitioner discussion of a near-identical case. An Eng-Tips thread covers an industrial/office building converted to basketball courts, with **11 columns removed** and a "maximum new girder span of 90-ft, and tributary roof width 47-ft". That is almost our geometry. The concerns raised were ponding, camber shedding water into adjacent bays, and settlement. The suggested fix was "9' or 10' deep trusses over the existing girders", with the structure hung from them (Eng-Tips).

Counterpoints. These projects are routine, not cheap:

- In a Queens factory church, a transfer structure "would have been too expensive in this building", so masonry walls were used instead (AISC).
- At the Fort Wayne coliseum, long-span trusses over the existing roof were rejected as "cost and schedule prohibitive" (Modern Steel, Feb 2003).
- An entertainment operator backfilling big boxes says "Columns and ceiling heights are usually the big issues" (REBusinessOnline, 2018).

Gap: I found no documented example that was specifically a **banquet or wedding hall** with columns removed. The documented precedents are churches, rinks and recreation centers, which are the same structural problem.

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3. Cost research and build-up

3.1 Published per-column removal costs

- **I found no published per-column cost** for removing a roof column in a single-story commercial building. None of the cost guides, bid tabulations or permit records I found covered it; residential "remove a basement column" guides don't apply.
- **Closest documented cost:** Crossgates Mall (2014). Removing 2 columns cost **$413k as post-tensioned concrete, about $207k per column**. The steel option was **$957k–$963k, about $480k per column**. That job carried about 1,000 kips from a 4-story building, roughly 10x our loads, but over a shorter 75-ft span (ProCM).

- **Tonnage benchmark:** Tulsa used 510 tons of renovation steel to remove 36 columns, 18 of them roof columns. The article doesn't separate out the column-removal steel (MSC June 2024).

3.2 Structural steel cost per ton, 2024–2026

Date / source	Supply (fabricated) $/ton	Erection $/ton	Installed $/ton	Note
Nov 2024, ConstructConnect via Linbeck Q4 2024	$3,500 (Houston) / $3,700 (N. TX) / $3,900 (Central TX)	$1,618 / $2,000 / $1,707	**$5,118–$5,700 [calc: sum]**	New construction, Texas metros
Nov 2025, Schuff Steel "The Steel Report" via Linbeck Q4 2025	$4,167 / $4,138 / $4,288	$1,916 / $1,925 / $1,802	**$6,063–$6,090 [calc]**	New construction
Schuff 12-month forecast (late 2026), same source	$4,306–$4,485	$1,893–$2,011	**≈ $6,309–$6,378 [calc]**	+3.7–4.6% supply, +3.9–5.0% erection
RSMeans national structural-steel price (Gordian)	Jan 2026 $2,343.93 → Oct 2026 **$2,721.31/ton** (+9.85% year over year)	–	–	Gordian doesn't define the figure; its size suggests material only. Use it as a trend indicator (Construction Executive, Apr 2026; Dec 2025)
Jan 2026 (NYREJ)	"recent increases of $35/ton on wide flange steel and $60/ton on plate steel"	–	–	–

- **Memphis pricing:** I found no published Memphis figure. Texas is the nearest published benchmark, and local fabricator quotes are needed.
- **Retrofit premium:** I found **no published source** for how much more retrofit steel costs per ton than new construction. The rates used in §3.8 are **[assumption]**: **$8,000 / $10,000 / $12,000–12,500 per ton**, about 1.25x / 1.6x / 1.9x the new-build rate. The premium covers small tonnage, single 89-ft members (oversize shipping or a field splice), erection inside an existing building, field welding to 1984 steel, and coordination with shoring.
- **Joist girders:** I found **no public $/ton price**. Gordian reports that open-web steel joists "spiked roughly 12% nationally" (Construction Executive, Apr 2026). SJI 100-2025 sets the product limits: joist girders are standardized at depths of 20–120 in. and spans of 20–120 ft, and the span "shall not exceed 24 times the depth" (SJI 100-2025). For an 89-ft span, that means a depth of at least 44.5 in. **[calc]**.

3.3 Temporary shoring and jacking

- Rental benchmarks (Projul, 2025), which cite no underlying source:
  - Heavy-duty shoring frames: $15–$40 per frame per month
  - Hydraulic jacks with needle beams: $25–$75 per jack
  - Proprietary engineered towers: **$50–$200 per tower per day**
  - Shoring engineering: **$2,000–$15,000**
- Method guidance:

- Preload the transfer member with jacks before cutting the column, so that "only the transfer girder undergoes deflection and the supported floors are unaffected" (STRUCTURE, preloading approach).
- "Jacks should be used to preload the system" before the existing support is released (STRUCTURE, Meehan 2004).
- Tulsa jacked its trusses 2 to 3½ in. (MSC).

3.4 Footings at the girder ends

- Unit prices from Linbeck Q4 2025, Texas:
  - 4,000-psi concrete: **$150–$178/CY**
  - Rebar supply: **$1,192–$1,334/ton**, plus install **$473–$540/ton**
- On a small interior footing, materials are a small share. Labor and access dominate: saw-cutting the slab, hand or mini-excavator digging, doweling into the old footing, patching the slab. Per-footing costs in §3.8 are **[assumption]**.
- Precedents:
  - Tulsa replaced 18 footings (MSC).
  - Boulder 29 enlarged footings at the first interior column line and stitch-welded plates onto overloaded W8 columns (MSC Jan 2025).

3.5 Engineering fees

- **Benchmark:** the State of Tennessee designer-fee formula is fee % = 35 / ($\log_{10}$(construction cost) − 1.15).
  - The State's renovation example applies a 1.25 multiplier: $1,000,000 × 7.216% × 1.25 = **$90,206 (about 9.0%)** (University of Memphis Designer Manual, Ch. 2).
  - The same formula gives **9.8% at $400k and 8.9% at $1.1M [calc]**.
  - These are full architecture/engineering fees on State work.
- I carry 8% / 10% / 12% for structural design, field verification, permit support and construction administration **[assumption, benchmarked to the above]**.

3.6 Special inspections, geotechnical work, roofing, MEP

I found no per-item published costs. These are carried as **[assumption]** allowances in §3.8.

3.7 Steel tonnage per removed column (from §4)

- **Low:** a deep joist girder (about 4–5 t), plus seats, bracing and end-column reinforcing, ≈ **6.5 t**
- **Mid:** a W40×215-class girder (9.6 t), plus 20% for connections, stiffeners, bracing and column reinforcing, ≈ **11.5 t**
- **High:** a stiffness- or ponding-driven W44×335-class girder or an above-roof truss, ≈ **15 t**

3.8 Cost build-up per removed column

Line item	Basis	Low	Mid	High
Transfer steel (girder, connections, bracing, end-column reinforcing)	Tons from §3.7 × retrofit $/ton (§3.2)	6.5 t × $8,000 = **$52,000**	11.5 t × $10,000 = **$115,000**	15 t × $12,000 = **$180,000**

Line item	Basis	Low	Mid	High
Temporary shoring, jacking and preload, incl. shoring engineering	Projul ranges + labor **[assumption]**	$8,000	$15,000	$25,000
Footings at the 2 girder-end columns (enlarge or new)	Linbeck unit prices + labor **[assumption]**	$6,000	$25,000	$40,000
Geotechnical work / test pits (share)	**[assumption]**	$1,500	$3,000	$5,000
Cut and remove the column and its 24-in. cover, patch slab	**[assumption]**	$3,000	$5,000	$8,000
Roof opening for the lift, patch, drainage adjustments	**[assumption]**	$3,000	$12,000	$25,000
Relocate MEP and sprinklers along the girder	**[assumption]**	$3,000	$10,000	$20,000
Fire protection of new steel	$0 if the building is Type IIB (§6.5)	$0	$0	$8,000
Special inspection and testing	**[assumption]**	$2,000	$4,000	$6,000
Direct cost		**$78,500**	**$189,000**	**$317,000**
GC general conditions and OH&P	12% / 15% / 18% **[assumption]**	$9,420	$28,350	$57,060
Structural engineering and construction administration	8% / 10% / 10% (benchmark §3.5)	$7,030	$21,740	$37,410
Contingency (framing type still unknown)	10% / 15% / 20% **[assumption]**	$9,500	$35,860	$82,290
All-in per column		**≈ $105,000**	**≈ $275,000**	**≈ $495,000**
× 4 columns		**≈ $418,000**	**≈ $1,100,000**	**≈ $1,975,000**
Hall A only (2 columns)		**≈ $209,000**	**≈ $550,000**	**≈ $988,000**
Owner-managed at Memphis prices (feasibility budget): direct cost × 0.89, no GC general conditions or OH&P, engineering and contingency kept		**≈ $85,000**	**≈ $175,000**	**≈ $300,000**

What moves the cost toward Low:

- A deep joist girder fits above the planned ceiling.
- The roof is shown not to be ponding-susceptible.
- The existing end footings have reserve capacity.
- The girder can be brought in through a door and erected in pieces.
- All columns are done in one mobilization.

What moves it toward High:

- Ponding or headroom forces a heavy or above-roof member.
- New footings or underpinning are needed.

- A girder end lands on an exterior-wall column or on a column shared by Hall A and Hall B (load roughly doubles there; §6.2).
- Sprinkler and roof-drain rework is needed.
- Fireproofing is required.

Cross-check against research:

- The Mid all-in figure ($275k) falls between Crossgates' post-tensioned ($207k) and steel (~$480k) per-column costs. Crossgates had about 10x the load but a shorter span, and was priced in 2014.
- Our steel tonnage of 26–60 t for four columns **[calc]** is consistent with Tulsa's 510 t for a much larger 36-column job.

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4. Sanity check (my calculations) [calc]

Geometry.

- Grid: 44.5 ft x 44.5 ft. Tributary area of each removed column: **At = 1,980 SF**.
- New girder span: **L = 89 ft** (two bays), running across each hall's 83.5-ft width between the columns on the long walls.

The girder can be oriented two ways:

- **Option A:** along the existing girder line, carrying the joists as a uniform load.
- **Option B:** parallel to the joists, picking up the girder line as a point load at midspan.

Both give the same midspan moment, because $P \cdot L/4 = w \cdot L^2/8$ when $P = w \cdot L/2$. The end-column reactions are also the same.

4.1 Loads

- **Dead load:** 15–20 psf (given).
- **Roof live load:** 20 psf. The IBC allows reduction to $Lr = Lo \cdot R1 \cdot R2$, with **R1 = 0.6 for At ≥ 600 SF** and $12 \le Lr \le 20$ psf. For At = 1,980 SF, that gives **Lr = 12 psf** (IBC model text, Eq. 16-26 to 16-29, as printed in the Seattle 2015 code). I bracket the range: unreduced 20 psf (conservative, often preferred for a public-assembly transfer member) and reduced 12 psf.
- **Snow:** I could not retrieve the ASCE 7 site value. NOAA's Engineering Weather Data for Memphis lists a **50-year ground snow load of 6 psf** (NOAA). The USDA Forest Service lists Tennessee as "10 psf–25 psf," with higher values at elevation, and says to contact local officials (USDA FS). ASCE 7-22 snow loads come from the ASCE Hazard Tool and depend on risk category (Nucor Building Systems). At pg ≈ 6–10 psf, balanced roof snow is below the 20-psf roof live load. Drifts at new RTUs and parapets still need to be checked.

4.2 Point load, moment, shear

Case	Service P	Factored Pu (1.2D + 1.6Lr)	Mu incl. self-weight	Zx required (Fy = 50, φ = 0.9, braced)
High: D = 20, Lr = 20	79.2 k	110.9 k	2,705–2,776 k-ft	721–740 in^3
Low: D = 15, Lr = 12	53.5 k	73.7 k	1,877–1,948 k-ft	500–519 in^3

Shear Vu is about 48–69 k. Shear does not govern.

4.3 Stiffness requirements

- **Live-load deflection.** SJI 100-2025 allows L/240 for roofs, or L/360 where a plaster ceiling is attached (SJI 100-2025 §5.10).
  - At L/360 (2.97 in.), Ix must be at least **11,700 in⁴** for the unreduced live load, or 7,000 in⁴ for the reduced load.
- **Total-load deflection without camber.** At L/240, Ix must be at least about **18,000 in⁴**. In practice, dead-load deflection is handled with **camber or jack preload**; SJI's standard camber at 90 ft is about 3½ in.
- **Ponding** is likely to govern.
  - IBC 2021 §1611.2: "Susceptible bays of roofs shall be evaluated for ponding instability in accordance with Chapters 7 and 8 of ASCE 7" (STRUCTURE, 2021 IBC changes).
  - SJI 100 puts the ponding investigation on the specifying engineer and points to AISC/SJI Design Guide 40.
  - I ran the AISC 360 Appendix 2 simplified check, $C_p = 32 \cdot L_s \cdot L_p^4 / (10^7 \cdot I_p)$, with Lp = 89 ft and Ls = 44.5 ft. It needs **Ip ≈ 35,700 in⁴** to reach Cp ≤ 0.25, even before the joists' contribution. That is about **3x what strength needs**. (The formula is from the AISC Specification; I did not verify it against a link in this session.)
  - For comparison, the existing 44.5-ft girders give Cp = 558/Ip, so doubling the span raises the ponding term **16x**.
  - Unless the engineer shows the bay is not "susceptible" (adequate slope and working drains) or uses the refined method, ponding stiffness will set the member size.

4.4 Candidate sections

Section properties are approximate values from the AISC Manual, from my own knowledge, not checked against a link; the engineer must verify them.

Section	Weight per 89-ft girder	Strength ratio (high case)	LL deflection (20 psf)	Total deflection, no camber	Ponding Cp (simplified)
W40×167	7.4 t	1.03 (NG unless Lr is reduced)	2.99 in. (L/357)	6.7 in.	0.77
W40×199	**8.9 t**	0.83	2.33 in. (L/459)	5.3 in.	0.60
W40×215	**9.6 t**	0.75	2.08 in. (L/515)	4.8 in.	0.53
W44×262	**11.7 t**	0.58	1.44 in. (L/743)	3.4 in.	0.37
W44×335	14.9 t	0.47	1.11 in. (L/958)	2.8 in.	0.29

Result:

- Strength plus live-load deflection with camber: a **W40×199 to W44×262 (about 200–260 lb/ft, 39–44 in. deep)**, or **8.9–11.7 t per girder**.
- Adding 15–25% for connections, stiffeners, bracing and end-column reinforcing gives about **10–14.6 t per removed column**.
- If ponding governs with a W-shape, expect **15 t or more**, or switch to a deep truss.

4.5 Joist-girder alternative

I sized this from chord forces, with service M ≈ 1,860 k-ft, chords at 0.6Fy, and webs/gussets adding 30–50%.

Depth	Weight	Steel per girder	I_eff	LL deflection	Cp
72 in.	~97–112 plf	4.3–5.0 t	~21,500 in⁴	L/663	0.42
84 in.	~82–95 plf	3.7–4.2 t	~25,300 in⁴	L/780	0.35
96 in.	~72–83 plf	3.2–3.7 t	~29,100 in⁴	L/897	0.31

The joist girder is about half the steel of a W-shape and stiffer, but **6–8 ft deep**. A joist-girder top chord counts as braced only where joists bear on it with positive attachment (SJI 100 §4.4.1). In Option B, almost nothing bears on it, so bracing must be added.

4.6 Tons to cost, compared with research

- **W-shape route:** 10–14.6 t × $8,000–$12,500/t ≈ **$80k–$183k of steel per column**.
- **Joist-girder route:** 4.5–6.5 t × $8,000–$10,000/t ≈ **$36k–$65k**.
- These bracket the steel line in §3.8.
- The all-in results (§3.8) are consistent with the only documented per-column cost (Crossgates, $207k–$480k per column at about 10x our load) and with Tulsa's tonnage.

4.7 End-column load increase

- Each girder-end column goes from **79.2 k to about 128.6 k** (unreduced case) or from **53.5 k to about 90 k** (reduced case). That is **+62% to +68%** service load, including girder self-weight.
- Two situations roughly double the load:
  - A girder end on an exterior-wall column (originally about half a bay of tributary).
  - A column on a line shared by Hall A's and Hall B's girders.

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5. Schedule

Step	Duration	Basis
Pre-purchase structural walkthrough	1 site day; report "within a few days to a week"	RSB Environmental
Existing-conditions survey after purchase (drawing search, joist tags, measurements, test pits)	2–4 weeks	**[assumption]**. SJI notes joist-tag data is "typically useful if the original project was constructed after 2000," so for this 1984 building the original drawings matter (SJI)
Structural design and permit drawings	3–8 weeks	Crossgates: "3 weeks to design" a 2-column removal (ProCM); upper end is **[assumption]**
Shelby County plan review	Average "7 to 20 working days" per cycle, plus resubmittals	Shelby County, General Information about Submittals. This is an older document (2015-IBC era); confirm current times.
Steel shop drawings and approval	+2 to 6 weeks	Terrapin, July 2026
Steel fabrication lead time	14–20 weeks nationally; Southeast 12–18; Gulf Coast 8–14. **40–50 weeks in some markets**, longest near large data-center campuses. 14–18 weeks (Jan 2026).	Terrapin; Skanska, Summer 2026; BD+C, Sept 2026; NYREJ
Joist-girder lead time	**Not found**	–

Step	Duration	Basis
Footings at the girder ends	About 1–2 weeks per pair plus cure; can run during steel fabrication	**[assumption]**
Field installation (shore, erect, connect, reinforce, preload, cut, remove shores)	About 2–3 weeks per column; 6–10 weeks for four with overlap; 3–5 weeks for Hall A. Crossgates took "2 months to implement in the field" for 2 heavily loaded columns.	ProCM; **[assumption]**
New RTUs for the halls (in parallel)	RTU lead times rising; major HVAC equipment up to 52 weeks; 2026 HVAC prices +10–12%	Skanska
Total	**About 6–9 months typical; 12+ months in a tight steel market**	For scale, Tulsa (36 columns plus a full rink fit-out) began truss erection about 8 months after design started and opened about 18 months after kickoff (MSC)

Timing note. The building is empty now. Book the walkthrough in §8 early, so its findings are in hand before any binding commitment.

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6. Design issues

6.1 Headroom and existing clear height

- **Headroom lost if the girder hangs below the existing structure [calc]:**
  - W40/W44: member depth about 39–44 in., plus about 6 in. for connections and clearance, so **about 3.7–4.2 ft lost**.
  - Joist girder: **6–8 ft lost**.
  - Option A, with the new member set beside or within the depth of the existing girder line: **0–2 ft**.
  - Above-roof truss: **0 ft**.
- **Typical existing clear heights:**
  - A worship-facilities builder: "converting a grocery store that already has 18-20 foot ceilings" (Worship Facilities, 2017).
  - A vendor-sponsored piece: "A masonry building built 50 years ago might have a maximum clear height of only around 14 feet to 16 feet" (Bisnow / Rooflifters).
  - I found **no authoritative survey of 1984 big-box clear heights**. Measure on site.
- **Implication.** At 16–20 ft to the bottom of the existing structure, a W40 hung below leaves about 12–16 ft under the girder line. That is why the girder's location within the structure, or an above-roof truss, matters as much as its size.

6.2 Can the end columns and footings take the added load?

- Expect **+62–68%** at each girder-end column (§4.7). It is roughly **+100%** where a girder lands on an exterior-wall column or a shared Hall A / Hall B column.
- **Practitioner view:** removing a column effectively doubles the outer columns' tributary area, and "It's unlikely there is that much 'fat' in the original design" (Eng-Tips).
- **Code:**

  - Under the IEBC "5% rule," existing gravity elements whose design loads rise by more than 5% must be altered or replaced to meet the current IBC (STRUCTURE, "The Basics of the 5% Rule", which covers the 2018 IEBC §502.4 and the similar §503.3 for alterations).
  - Shelby County has adopted the 2021 IEBC with local amendments (Shelby County 2021 IBC with Local Amendments).
- **Plan for:**
  - Column reinforcing (welded plates, as at Boulder 29).
  - Enlarged or new footings (Tulsa replaced 18).
  - A check of settlement under the higher soil pressure (Eng-Tips).
- **Get first:** the 1984 structural drawings and geotechnical report, if they exist.

6.3 Roof ponding, drainage, camber, and RTU loads

- **Ponding.** An 89-ft member under a flat roof is the classic ponding-sensitive case (§4.3). The old column point becomes the girder's midspan, the point of maximum deflection. Unless the girder is cambered or preloaded, that point becomes a new low point on the roof.
- **Drainage.** Check roof slope (structural vs. tapered insulation), drain and overflow locations, and secondary drainage. If reroofing is due (the roof dates from 1984 or the 1996 renovation; age unknown), combine it with girder installation and a re-slope.
- **RTUs.** Two 11,000-SF assembly rooms will need larger cooling units than a call center. Per SJI's retrofit panel, changing out rooftop mechanical equipment is the most common reason joists get retrofitted (SJI). Place new units over columns or on new framing, not at midspan of the new long-span bays, and check snow drift around them. RTU lead times are rising (§5).
- **Lateral system.** Confirm the interior columns are gravity-only. If joist-girder bottom chords or moment connections tie into them, removing them affects the lateral load path (see the Eng-Tips caution on lateral restraint). Tulsa had to rework its diaphragm chords, collectors and braced frames (MSC).

6.4 Code triggers from the change to assembly occupancy (independent of column removal, but structural)

- **Risk Category.** The 2021 IBC (adopted in Shelby County) assigns **Risk Category III** to buildings with assembly spaces "each having an occupant load greater than 300 and a cumulative occupant load ... greater than 2,500" (STRUCTURE, 2021 IBC changes; Building Code Forum).
- **Where the halls land [calc]:**
  - At 15 net SF per occupant (tables and chairs), the two halls total about 1,470 occupants, which stays Risk Category II.
  - At 7 net SF (chairs only, e.g. ceremony seating), they total about 3,140, which is Risk Category III.
  - The other planned assembly spaces (children's entertainment center, restaurants) also count toward the 2,500 if each exceeds 300 occupants.
  - The occupant-load factors are IBC Table 1004.5 values from my own knowledge, not checked against a link.
- **IEBC seismic requirement.** If the change of occupancy moves the building to a higher Risk Category, IEBC §1006.3 requires the building to meet the IBC seismic requirements for the new category, with limited exceptions (Hawaii's adoption of the 2018 IEBC §1006.3; model 2021 text, shown struck in Seattle's code).
  - Exception: SDS < 0.33.

  - Exception: new occupancy area < 10% of the building. 22,000 SF of hall area is about 17% of 127,634 SF **[calc]**, so this one doesn't apply.
- **Memphis seismicity.** Memphis has "a dense urban population near faults capable of producing major earthquakes," with a 25–40% chance of an M6.0+ in 50 years (USGS). I could not retrieve the parcel's SDS; the engineer must get it from the ASCE Hazard Tool.
- **This could cost more than the column work.** Ask about it at the pre-application meeting. Capping posted occupant loads may keep the building in Risk Category II.

6.5 Fire protection of new steel

- Under IBC Table 601, **Type IIB** construction requires **0 hours** for both the primary structural frame and the roof framing, so the new girders need no fireproofing. Type IIA requires 1 hour (model IBC Table 601, as printed in the Seattle 2012 code). Confirm the building's construction type from the permit record.
- **Related flag for the code consultant.** A-2 occupancies inside a one-story "unlimited area" building come with conditions: separation from other occupancies, a per-space area limit, and "exit doors from Group A-1 and A-2 occupancies shall discharge directly to the exterior of the building." These conditions were discussed for a Type IIB strip center (Building Code Forum; 2009-IBC numbering). They affect where Hall A and Hall B can go.

6.6 Memphis / Shelby County permit requirements

- **Code in force.** Shelby County Code Enforcement enforces the **2021 IBC with local amendments** in Memphis and unincorporated Shelby County, effective December 31, 2021. The design professional "shall be an architect or engineer legally registered under the laws of this state" (§107.3.4.2). The amendment list I reviewed shows no changes to Chapters 16 or 17 (Shelby County 2021 IBC with Local Amendments).
- **Submittals.**
  - Plans must be "signed and sealed by a licensed design professional in the State of Tennessee" for **Assembly** occupancies and for any space over 5,000 SF.
  - The package must include a "statement of special inspections if required" and structural plans "as needed" (Shelby County, Requirements for General Submittal Documents).
  - The older County guide also requires "an engineer's letter or structural calculations" for "any structural reconfiguration." It lists plan-review fees (for example, $875 for a $200,001–$500,000 project) and a permit fee of $100 plus $3.50 per $1,000 (older County guide). Confirm current fees.
- **State law.** For Assembly occupancies, "a registered design professional is required … regardless of the size of the facility" (Tennessee Board of Architectural & Engineering Examiners).
- **Special inspections.** Expect inspections under IBC Chapter 17 for structural steel welding and bolting, footing concrete, and soils, listed in the Statement of Special Inspections. These are carried in §3.8.

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7. Alternatives

Option	Pros	Cons	Precedent
A. Long-span joist girder below the roof (72–96 in. deep)	About half the steel of a W-shape (~3–5 t; §4.5); stiff, which helps with ponding; a standard SJI product (≤120 ft span, ≥44.5 in. deep for 89 ft); the manufacturer designs it	**6–8 ft headroom loss** if hung below; top-chord bracing needed; joist $/ton and lead time not found; custom connections to 1984 steel	Faith Family Church: joists and joist girders between 160-ft trusses (SJI)

Option	Pros	Cons	Precedent
B. Wide-flange transfer girder (W40–W44)	Shallower (~3.3–3.7 ft); any fabricator can make it; simple connections	9–15 t per girder; 89-ft piece means oversize shipping or a field splice; ponding may push it heavy	Crossgates (ProCM); Legacy Church jack beams (BGW)
C. Truss above the roof, roof hung from it	**No headroom loss**; very stiff and can be 8–12 ft deep; can be combined with a reroof	Roof penetrations and flashing at every post and hanger; exposed steel (corrosion, thermal movement); snow drift and wind; visible from outside; complex erection. At Fort Wayne, trusses over the roof were "cost and schedule prohibitive"	Heights Church (L.F. Jennings); Eng-Tips 9–10 ft trusses over girders (Eng-Tips)
D. Under-slung king-post truss reusing existing framing	Least new steel; reuses the existing girders and the column stub	Rods and post sit below the girder line, cutting headroom at mid-room; existing girders must work as compression chords; needs tensioning and jacking	Chelsea Piers CT (WSP); Tulsa (MSC)
E. Keep the columns (feature them using the existing 24-in. round covers)	$0 structural cost and no structural schedule; avoids most §6.2–6.3 risk	Sightlines, layout of the head table and dance floor, and marketability against column-free venues ("Columns and ceiling heights are usually the big issues", REBusinessOnline)	–
F. Relocate or reshape the halls	No structural cost	On a 44.5-ft grid, any 11,000-SF room will contain columns. Moving the room only improves where the columns fall (into walls, edges, or a prefunction line); it can't eliminate them. Exit-to-exterior rules (§6.5) also constrain location.	–

My call: carry **Option B or A (Mid case)** in the pro forma. Evaluate **Option C** if the walkthrough shows a low existing clear height or a roof due for replacement. Keep **Option E** as the fallback if the §6.4 code triggers or the footing findings come back adverse.

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8. Pre-purchase structural walkthrough: checklist and cost

What the structural engineer should check on site:

1. **Roof framing type and direction.**
   - Joists on joist girders, or wide-flange beams?
   - Which way do the girders run relative to Hall A and Hall B? This decides Option A vs. B.
   - Member depths and spacing.
   - Photograph the joist tags (manufacturer, job number, mark). SJI says tags can unlock the manufacturer's capacity records, and recommends the SJI Joist Investigation Form (SJI).
2. **Columns.** Size and shape, cap and base connections, any bracing or moment connections (are they part of the lateral system?), and condition.
3. **Clear height.** Slab to the bottom of structure at the four column lines; existing ceiling height; space for a 40–96 in. member; locations of sprinkler mains and ducts.
4. **Roof slope and drainage.** Slope direction, drain and overflow locations near the four columns, signs of past ponding (stains, debris lines, sagging bays), roof age and condition.

5. **Existing loads.** RTU sizes, locations and curbs; loads hung from the 1997 call-center fit-out.
6. **Foundations.** Slab cracking or settlement at the columns. Ask the seller and the County for the **1984 structural drawings and geotechnical report** and the full 1997 Sitel set.
7. **Lateral system and construction type.** Wall type, bracing, deck attachment, and whether the steel is unprotected (Type IIB?). These feed §6.4 and §6.5.
8. **Fire protection.** Sprinkler coverage and main routing.

Typical cost:

- **Visual structural assessment:** "$0.015/sf to $0.025/sf" per a practitioner (Eng-Tips). For 127,634 SF that is about **$1,900–$3,200 [calc]**, with calculations extra.
- **Full ASTM E2018 property condition assessment, 50,000–200,000 SF: $5,000–$10,000** (LegalClarity, July 2026, citing M2E Consulting Engineers).
- **Structural commercial inspection, 10,000–50,000 SF:** $4,000–$10,000; "$15,000 or more" for large or specialized facilities (Level Engineering).
- **My recommendation:** budget **about $3,000–$7,500** for a focused one-day structural feasibility walkthrough with a short letter, given the expedited timing **[assumption based on the above]**.

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9. Gaps and what I could not find

- No published **per-column removal cost** for single-story commercial roofs.
- No public **joist-girder $/ton** or joist-girder lead time.
- No published **retrofit-vs-new $/ton premium**; the factors in §3.2 are assumptions.
- No Memphis-specific steel pricing; Texas figures were used.
- The **ASCE 7 site values for Memphis** (ground snow pg, SDS) were not retrieved; they need the ASCE Hazard Tool.
- No authoritative source on **typical 1984 big-box clear height**.
- Shelby County **review times and fees** come from an older County document.
- The AISC section properties and the ponding formula are quoted from the AISC Manual and Specification from my own knowledge, not checked against a link.
- My web-search allowance ran out partway through, so some leads (RSMeans renovation factors, banquet-hall-specific precedents) were not pursued. They can be picked up in a follow-up.

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10. Sources

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- Steel Cooling: Steel Costs Steadily Decline After Pandemic Price Shock, Construction Executive, Apr 3, 2026
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- Hawaii amendments to 2018 IEBC Chapter 10 (§1006.3), Justia
- The Basics of the 5% Rule, STRUCTURE (K. Wagner)
- Unlimited area buildings: mixed occupancy building with use group A-2, The Building Code Forum
- NOAA Engineering Weather Data: Memphis, TN
- USDA Forest Service: National Snow Load Information, state list
- Engineering Tips: Snow Load Changes for ASCE 7-22, Nucor Building Systems

- Memphis Urban Seismic Hazard Maps, USGS

Permits (Memphis / Shelby County / Tennessee)

- Shelby County: 2021 International Building Code with Local Amendments
- Shelby County: Requirements for General Submittal Documents
- Shelby County: General Information about Submittals (older)
- Tennessee Board of Architectural & Engineering Examiners: Requirements for Building Design

Schedule and lead times

- Commercial Construction Material Lead Times in 2026, Terrapin Construction Group
- Skanska 2026 Summer Construction Market Trends
- Long lead times on materials is creating new project delivery challenges, BD+C, Sept 2026

Clear height

- Bisnow Studio B: raising industrial roofs (sponsored by Rooflifters)

Pre-purchase inspection cost

- Structural condition assessment fee, Eng-Tips
- How Much Does a Commercial Building Inspection Cost?, LegalClarity, July 2026
- The Comprehensive Guide to Choosing a Structural Commercial Building Inspector, Level Engineering
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Public version: prices, fees and transaction terms are omitted. Everything here comes from public records, desk research and concept drawings; it is not legal, engineering or investment advice.